

MASS-ENG-024

Enterprise Analytics Engineering Specification

Version 1.0

Engineering Objective

This specification implements the constitutional capability responsible for enterprise analytics stewardship. Enterprise Analytics transforms enterprise activity into trusted, actionable intelligence while preserving constitutional ownership boundaries. It measures the enterprise; it does not operate the enterprise.

Engineering Preface

Enterprise Analytics is the convergence point for enterprise intelligence. Every constitutional capability publishes measurable outcomes into this subsystem. Enterprise Analytics does not create operational data; it validates, aggregates, correlates, and presents enterprise truth. Dashboards are products of Enterprise Analytics—not its purpose. Its purpose is to provide reliable intelligence for every decision maker in the ecosystem.

Purpose & Scope

Govern enterprise metrics, KPI stewardship, trend analysis, performance baselines, cross-department reporting, forecasting support, executive reporting, and enterprise measurement standards. Out of Scope: operational execution, financial processing, communications delivery, relationship stewardship, and AI orchestration.

Architecture

Core Components: Analytics Service, Metrics Registry, KPI Service, Reporting Service, Trend Analysis Service, Correlation Service, Dashboard Service, Benchmark Registry, Executive Reporting Service, Enterprise Intelligence Repository.

Engineering Requirements

Functional: collect enterprise metrics, normalize measurements, calculate KPIs, detect trends, compare benchmarks, publish executive reports, expose analytics APIs. Non-Functional: accurate, auditable, observable, scalable, extensible, constitutionally compliant.

Dependencies & Responsibilities

Consumes all Platform Foundation specifications and Enterprise Capability specifications through Communications. Owns enterprise measurement standards, KPI governance, trend analysis, benchmarking, and reporting. Does not own operational decisions, customer relationships, marketing strategy, finance, or workflow execution.

Interfaces & Example Flow

Interfaces: RegisterMetric(), PublishKPI(), GenerateReport(), EvaluateTrend(), CompareBenchmark(), ExportAnalytics(). Example Flow: Enterprise capabilities publish events → Enterprise Analytics validates and correlates metrics → Executive Intelligence consumes insights → Leadership makes informed decisions.

Claude Implementation Checklist

Implement metric registry, KPI engine, reporting services, analytics APIs, benchmark framework, observability integration, automated validation, and constitutional traceability.

Definition of Done

Enterprise Analytics provides a trusted, measurable, and constitutionally governed representation of enterprise performance that enables informed decision-making across the entire MASS HQ ecosystem.